● PRODUCTION VERIFIED • LIVE

Version: 2.0.0 | **Date:** Sep 2026
Status: Fully Operational
Deployment: Netlify + Render

Param Retrieval & Analytics Studio

Technical System Architecture & Executive Handover Report

1. Executive Summary & Problem Solved

Param Retrieval Studio is a cloud-native search and telemetry analytics platform. It indexes real visitor interaction logs (dwell durations, button presses, exhibit stations) from museum telemetry datasets (engagement-sessions.csv) across a hybrid vector engine.

The Core Innovation: Real Math Computation vs. Keyword Matching
Traditional search engines only match words. If an executive asks: *'What is the minimum, maximum, and average dwell duration across exhibits?'*, legacy search engines merely look for documents containing the literal words 'minimum' or 'average'.
Param performs authentic computation: It retrieves real telemetry data points across exhibits, extracts the numbers, computes actual statistical values (MIN, MAX, and arithmetic mean AVG), and returns verified answers with step-by-step mathematical formulas.

2. Step-by-Step Technical Lifecycle

1. Data Ingestion & Dual Indexing

Visitor telemetry logs are transformed into two complementary formats: 384-dimensional dense semantic vectors (via BAAI/bge-small-en-v1.5) for conceptual search, and sparse lexical tokens (via BM25) for exact exhibit IDs. Vectors are indexed in Qdrant Cloud; relational metadata is stored in Supabase PostgreSQL.

2. Strict Access Control & Whitelist Guard

User sessions are authenticated with Supabase Auth. Access is strictly restricted to 5 pre-approved team email addresses (e.g. mantriramitha.cs24@bmsce.ac.in, mantriramitha17@gmail.com). Unauthorized users are blocked at both frontend and API gateway levels.

3. Intelligent Query Formulation

When an analytical calculation query (e.g. 'what is the min and max dwell duration?') is submitted, the API gateway automatically strips calculation words ('min', 'max', 'average') before sending to BM25. This ensures the database searches for the actual entity and metric rather than matching literal keywords.

4. Hybrid RRF Retrieval & Cutoff Filtering

The retrieval engine queries dense vectors and sparse BM25 in parallel. Results are fused using Reciprocal Rank Fusion (RRF: 1/(60+Rank_dense) + 1/(60+Rank_lexical)). Chunks falling below the relevance score threshold (RRF < 0.012) are pruned to eliminate irrelevant noise.

5. Deterministic Math & Telemetry Compute

The compute engine inspects extracted telemetry attributes (duration_s, mean_blackout_duration_s, presses, saccades) across exhibits. It calculates true MIN, MAX, and arithmetic mean AVG (Sum ÷ Count), formatting the exact formula and exhibit associations without hallucination.

6. Multi-Model Gemini RAG with Zero-Error Fallback

Retrieved evidence is synthesized using Google Gemini Flash under strict grounding rules ('Searching words is NOT data analysis'). If remote AI APIs experience high-demand spikes (503), the engine automatically falls back to deterministic compute for 100% uninterrupted uptime.

7. Clean Presentation Studio (Data Over Files)

Built with React 19. For analytical queries, the UI hides raw file cards (#1 file.csv) to avoid screen clutter. Instead, it displays the AI Answer Card and the Extracted Telemetry Data Table (Exhibit Station, Dwell Duration, Interactions, and UTC Timestamps) with a toggle to inspect source chunks.

3. Key Engineering Highlights

■ Zero-Crash Architecture Built-in deterministic compute fallback eliminates 503 LLM outage errors and ensures the user always gets verified numerical answers.	**■ Multi-Layer Security** Strict 5-email whitelist guard, 60 req/min IP rate limiting, server-side secret token isolation, and preflight CORS protection.
■ Clean Data Presentation Automatic mode switching prioritizes structured telemetry data tables over raw source file chunks for all question and calculation queries.	**■ Hybrid RRF Precision** Combines deep semantic vectors with high-precision BM25 lexical token matching to yield top relevance scores across museum telemetry.

4. Production Deployments & Cloud Topology

Component	Hosting Platform	Technology	Live Endpoint / Identifier
Frontend Studio UI	Netlify CDN	React 19, Vite	`https://param-search-studio.netlify.app`
API Gateway	Render Web Service	Node.js, Express	`https://param-retrieval-backend.onrender.com`
Retrieval Engine	Render Service	Python, FastAPI	`https://param-retrieval.onrender.com`
Vector Database	Qdrant Cloud	Dense + Sparse Index	Index ID: ddedb6e5febc4c78d93d
Auth & Database	Supabase Cloud	PostgreSQL, GoTrue	adtxcydewzfxszfuhhro.supabase.co
Code Repositories	GitHub	Git Source Control	`https://github.com/Rams1700/Param`